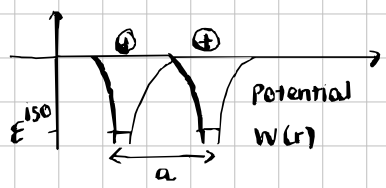
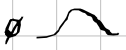
Electrons in Crystal

In Hydrogen Atom

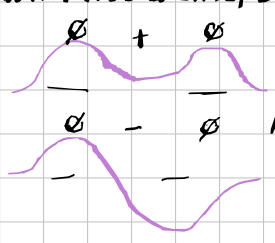


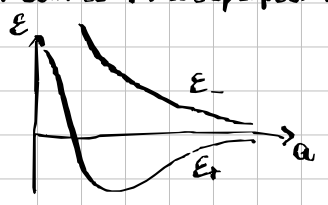
It has the Hamiltonian of

$$\hat{H} = \frac{\hat{p}^2}{2m} + W(\vec{r})$$
$$= -\frac{\hbar^2}{2m}\nabla^2 + W(\vec{r})$$

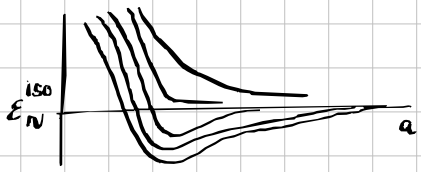
with ground state ϕ 

With two atoms, one electron can be in a superposition



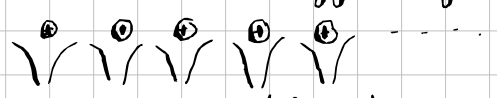


The superposition phase for this one electron creates different energy levels



At certain a, degeneracy is lifted at most N

But at $a \to \infty$, the energy is degenerate again



Even though, as $a \to \infty$ it seems like the electron should be localized for only one atom. That is correct except that think of it as you start with a superposion and seperate or you start with a "single" atom. This thinking gives

localized  or "localized" 

Consider this electron as free particle $\hat{H} = -\frac{\hbar^2}{2m}\nabla^2$. A plane wave $\psi_k(\vec{r}) \sim e^{i\vec{k}\cdot\vec{r}}$ $\vec{p} = \hbar\vec{k}$ $E(\vec{k}) = \frac{\hbar^2|\vec{k}|^2}{2m}$ is the eigenstate

$0 \to \vec{k}$



Note that as $\vec{k}\uparrow$, it does not repeat. Lattice allows the repetition

But in a lattice, there is also a potential

$$\left(-\frac{\hbar^2\nabla^2}{2m} + W(\vec{r})\right)\varphi(\vec{r}) = \mathcal{E}\varphi(\vec{r}) \quad \text{with } W(\vec{r}+\vec{R}_n) = W(\vec{r})$$

The solution becomes

$$\psi_{\nu\vec{k}}(\vec{r}) = \frac{1}{\sqrt{V}} e^{i\vec{k}\cdot\vec{r}} u_{\nu\vec{k}}(\vec{r}) \qquad u_{\nu\vec{k}}(\vec{r}+\vec{R}_n) = u_{\nu\vec{k}}(\vec{r})$$

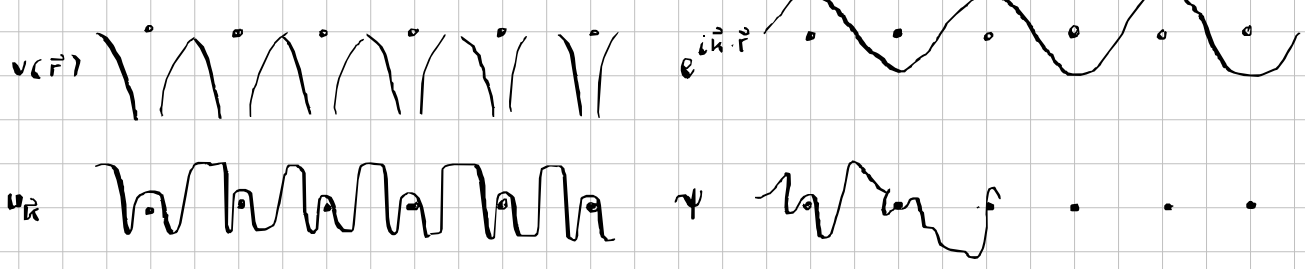
make G discrete

ν: band index / Energy level

The Bloch theorem says

$$\psi_{\nu\vec{k}}(\vec{r}+\vec{R}) = \psi_{\nu\vec{k}}(\vec{r})\, e^{i\vec{k}\cdot\vec{R}}$$

where $\vec{k}\cdot\vec{R}$ can be anything!



Thus for a general structure

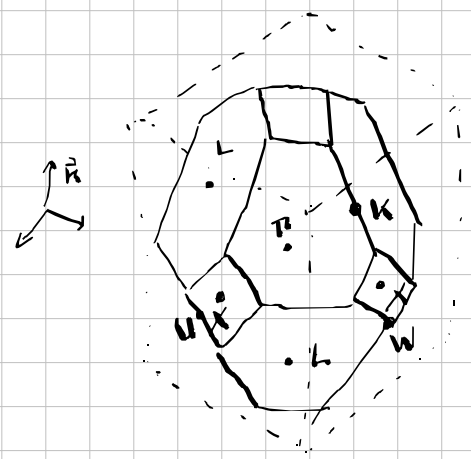
$$\left(-\frac{\hbar^2\nabla^2}{2m} + W(\vec{r})\right)\varphi_{\nu\vec{k}}(\vec{r}) = \mathcal{E}_{\nu\vec{k}}\,\varphi_{\nu\vec{k}}(\vec{r})$$

W: lattice potential $\quad \mathcal{E}_{\nu\vec{k}}$: energy band structure

$$\varphi_{\nu\vec{k}+\vec{G}}(\vec{r}) = \varphi_{\nu\vec{k}}(\vec{r}) \qquad \mathcal{E}_{\nu\vec{k}+\vec{G}} = \mathcal{E}_{\nu\vec{k}}$$

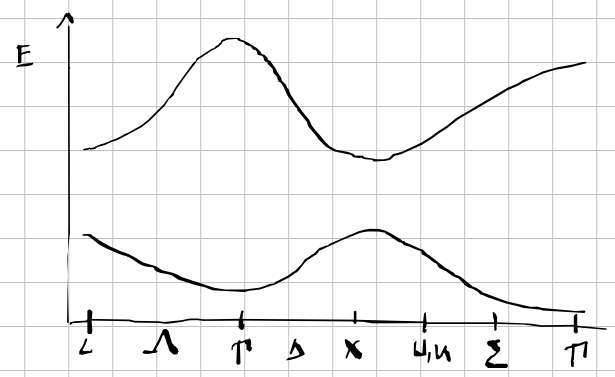
Even when we look at R_n away from last position or the electron wave vector is increased by G. The energy is the same.

Observe the reciprocal lattice of an FCC crystal (BCC reciprocal), We start with $\Gamma = \vec{k} = 0$



X is center of square face
L is center of hexagon face
W is vertex
K midpoint of edge between two hexagon
U midpoint of a square and hexagon
Δ: $\Gamma \to X$
Λ: $\Gamma \to L$
Q: $L \to W$
Σ: $X \to U$

When a band structure plot has a horizontal axis such as $\Gamma \to X \to W \to K \to L$, it means we calculate the energy along these directions. K in the x-axis is not magnitude



Caveats: The bloch theorem allows $\vec{k}\cdot\vec{R}$ to be any phase. This says that any electron with any $\vec{k}$ is allowed in the crystal. However a $\vec{k}+\vec{G}$ is reproducing the same wave function. G is discrete: allowed shift while $\vec{k}$ is continuous in an infinite crystal.

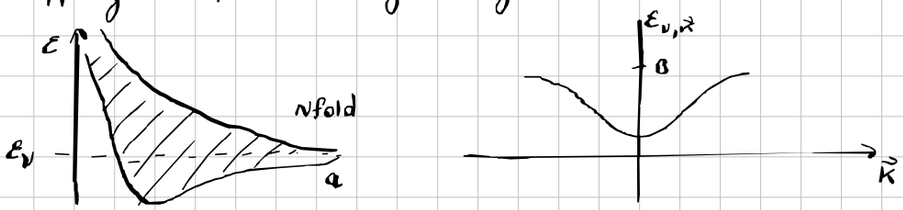
Tight Binding Model

Consider putting atoms closer such as there is an overlapping with wave fuctions



The overlapping lifts up N fold degeneracy

This creates a continuos band



We will see how this energy band arrives, Consider the Schrödinger

$$\left\{-\frac{\hbar^2}{2m}\nabla^2 + W(\vec{r})\right\}\varphi_{\nu\vec{k}}(\vec{r}) = \varepsilon_{\nu\vec{k}}\,\varphi_{\nu\vec{k}}(\vec{r})$$

In one dimension model $\vec{r} \to x$

$$W(x) = \sum_{\ell=1}^{N} W_{ion}(x - R_\ell) \quad \text{with} \quad R_\ell = \ell \cdot a \qquad N \to \infty$$

The solution for each atom is

$$H^0 \varphi_\nu(x) = \varepsilon_\nu^0 \varphi_\nu(x) \quad \text{with} \quad H^0 = -\frac{\hbar^2\nabla^2}{2m} + W_{ion}(x)$$

Construct

$$\varphi_{\nu\vec{k}}(x) = \frac{1}{\sqrt{N}} \sum_{\ell} e^{ikR_\ell}\,\varphi_\nu(x - R_\ell) \qquad \varepsilon_{\nu k} = \frac{\int dx\, \varphi^*_{\nu k}(x)\, H\, \varphi_{\nu k}(x)}{\int dx\, \varphi^*_{\nu k}(x)\, \varphi_{\nu k}(x)}$$

This is a superposition of an electron with a phase e^{ikR_ℓ} This phase changes from one site to Other.

$$\varepsilon_{\nu k} = \frac{1}{N}\sum_{nm} e^{-ik(R_n - R_m)} \int dx\, \varphi_\nu^*(x - R_n)\left\{\varepsilon_\nu^0 + \Delta V_{\ell,m}(x)\right\}\varphi(x - R_m) \qquad \Delta V_{\ell,m} = \sum_{\ell \neq m} W_{ion}(x - R_\ell)$$

Where other nearby potentials near m contribute the perturbation

Assume small overlap

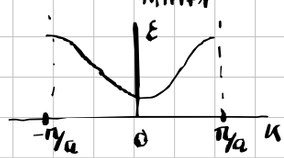


First order ↓ Secondorde ↙ ↘

$$\int dx\, \varphi_\nu^*(x - R_n)\,\Delta V_{\ell,m}(x)\,\varphi_\nu(x - R_m) \approx V_0\,\delta_{nm} + V_1\,\delta_{n,m+1} + V_{-1}^*\,\delta_{n,m-1}$$

$$\varepsilon_{\nu k} = \varepsilon_\nu^0 + V_0 + V_1 \frac{1}{N}\sum_{nm}\left(e^{-ik(R_{m+1} - R_m)}\,\delta_{n,m+1} + e^{-ik(R_{m-1} - R_m)}\,\delta_{n,m-1}\right) \qquad R_{m\pm1} - R_m = \pm a$$

$$= \varepsilon_\nu^0 + V_0 + 2V_1\cos(ka)$$



This perturbation theory predicts a cosine structure for the band

An electron near the bottom of a conduction band behaves approximatly like a free electron, but with an effective mass

At $T = k = 0$, expand taylor series

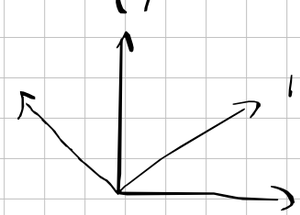
$$\varepsilon_{c,k} \approx \varepsilon_{c,0} + \frac{1}{2}\frac{\partial^2 \varepsilon_{c,k}}{\partial k^2}\bigg|_{k=0} k^2 = \varepsilon_{c,0} + \underbrace{\frac{\hbar^2 k^2}{2m_c}}_{\text{free energy}}$$

At minimum T $\quad \frac{\partial \varepsilon}{\partial k} = 0$

ε_{ck} is dispersion of free particle with effective mass m_c

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & \cdot \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \quad x' = \frac{1}{\sqrt{2}}x + \frac{1}{\sqrt{2}}y \qquad \varepsilon$$

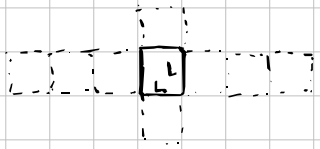
$$y' = \frac{-1}{\sqrt{2}}x + \frac{1}{\sqrt{2}}y$$





Periodic Boundary Condition

Until now, the volume is essentially infinite. But, in reality the volume $V = L^3$ is finite.



Infinite extension in each direction

If we make $\varphi(x+L) = \varphi(x)$ in all direction. This is mathematically the same as infinite crystal That is

$$e^{ik_x(x+L)} = e^{ik_x x} \Rightarrow k_x L = 2\pi$$

Finite crystal makes the wavelength finite.

$$k_x = n\frac{2\pi}{L} \qquad \Delta k_x = \frac{2\pi}{L} \qquad \Delta^3 k = \Delta k_x \Delta k_y \Delta k_z = \left(\frac{2\pi}{L}\right)^3$$

As $L \to \infty$, this k is continuous.

Using

$$\sum_{\vec{k}} = \frac{1}{\Delta^3 k} \sum \Delta^3 k \to \left(\frac{L}{2\pi}\right)^3 \int d^3k$$

Density of states can be evaluated

$$2 \sum_{\text{spin}} \frac{1}{V} \sum_{\vec{k}} f(\varepsilon_{\nu\vec{k}})$$

As $V \to \infty$, $\vec{k}$ is continuous, we ask how many states have energy near ε rather than what are all individual $\vec{k}$ states

Example is the density of electrons in a band

$$n_c = \frac{N_c}{V} = 2\frac{1}{V}\sum_{\vec{k}} f(\varepsilon_{\nu\vec{k}}) \to \int d\varepsilon \, g_\nu(\varepsilon) f(\varepsilon)$$

$$g_\nu(\varepsilon) = \frac{\text{number of available states } \vec{k} \text{ around } \varepsilon \text{ in band } \nu}{(\text{unit volume})(\text{unit energy})}$$

Formally, one can write as how many $\vec{k}$ contain same energy

$$g_\nu(\varepsilon) = \frac{2}{V}\sum_{\vec{k}} \delta(\varepsilon - \varepsilon_k) \Rightarrow \int d\varepsilon \, g_\nu(\varepsilon) f(\varepsilon) = \frac{2}{V}\sum_{\vec{k}} F(\varepsilon_{\nu\vec{k}})$$

Since tight binding model gives

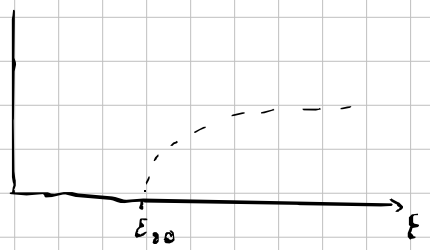
$$\varepsilon_{\nu\vec{k}} = \varepsilon_{\nu 0} + \frac{\hbar^2 k^2}{2m_\nu} \Rightarrow k = \frac{\sqrt{2m_\nu(\varepsilon - \varepsilon_{\nu 0})}}{\hbar}$$

The amount of k-states lie in thin spherical cell is

$$dN \propto 4\pi k^2 dk \qquad g_\nu(\varepsilon) \propto k^2 \frac{dk}{d\varepsilon}$$

$$\frac{d\varepsilon}{dk} = \frac{\hbar^2 k}{m_\nu} \qquad \frac{dk}{d\varepsilon} = \frac{m_\nu}{\hbar^2 k} \Rightarrow k^2 \frac{dk}{d\varepsilon} \propto m_\nu^{3/2}\sqrt{\varepsilon - \varepsilon_{\nu 0}}$$

$$g_\nu(\varepsilon) = \frac{1}{2\pi^2}\left(\frac{2m_\nu}{\hbar^2}\right)^{3/2}\sqrt{\varepsilon - \varepsilon_{\nu 0}} \;\; \Theta(\varepsilon - \varepsilon_{\nu 0})$$



Electron Distribution

Consider how likely each electron state in a band can be occupied

$$f_\nu(\vec{\kappa}) = \langle a^\dagger_{\nu\vec{\kappa}} a_{\nu\vec{\kappa}} \rangle = \frac{1}{e^{\beta[E_\nu(\vec{\kappa})-\mu]}+1}$$ ⇒ Lower energy allows better occupation

$\beta = 1/(k_B T)$

ν: band index $\vec{\kappa}$ is crystal vector and $E_\nu(\vec{\kappa})$ is the energy of that state μ is chemical potential

f indicates the average number of particles in state $\nu, \vec{\kappa}$

The chemical potential is chosen such that

$$N_e = \sum_{\nu,\vec{\kappa}} f_\nu(\vec{\kappa})$$

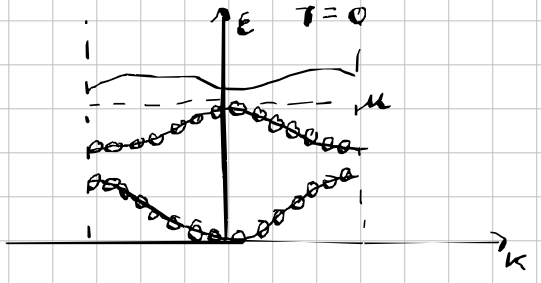
Particle density $n = \frac{N}{V}$

At limit $T \to 0$, if $E < \mu$, $e^{\beta(E-\mu)} \to 0$ $f_\nu(\vec{\kappa}) \to 1$

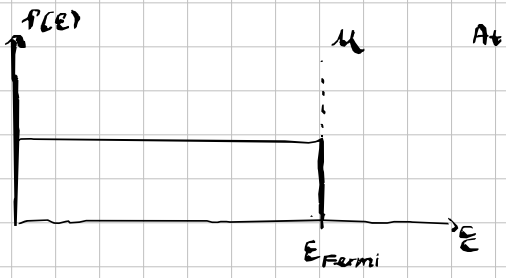
if $E > \mu$, $e^{\beta(E-\mu)} \to \infty$ $f_\nu(\vec{\kappa}) \to 0$

At zero temperature, there is no air particle kicking the electron so that it can only occupy states below μ

When temperature increases, some particles have more energy to occupy above μ

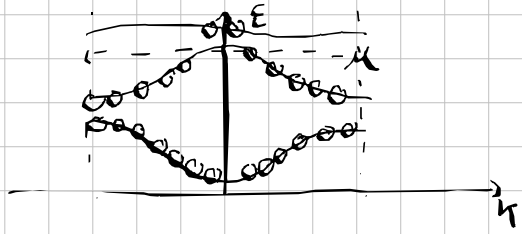


T=0



At $T=0$, $\mu = E_{Fermi}$

$T > 0$: Fermi edge

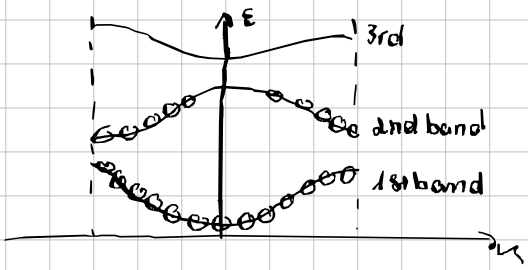


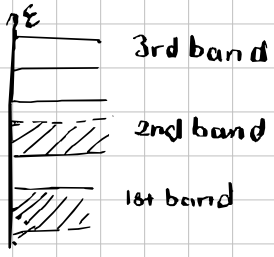


When T is large enough $\beta \to 0$, this is <u>maxwell distribution</u> $f(E) \to e^{-\beta(E-\mu)}$

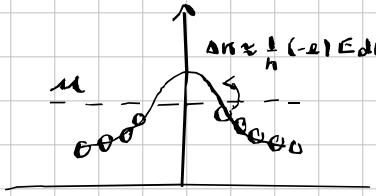
Classification of Solid

At Fermi distribution, $T \sim 0$ the typical scale for ε is eV since $k_B T \sim$ meV even at 300K



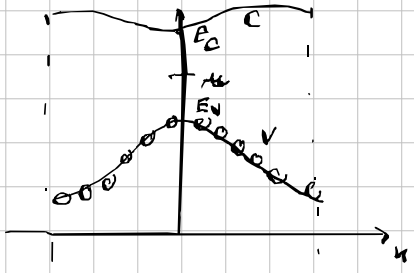


If μ is within a band, it is metal



An applied static E field can easily accelerate electrons in the band producing electrical current

If μ is between bands, it is insulator / semiconductor



The electrons in valance band create no current due to symmetry, but yet has no energy to excite to upper bound

Summary:

